

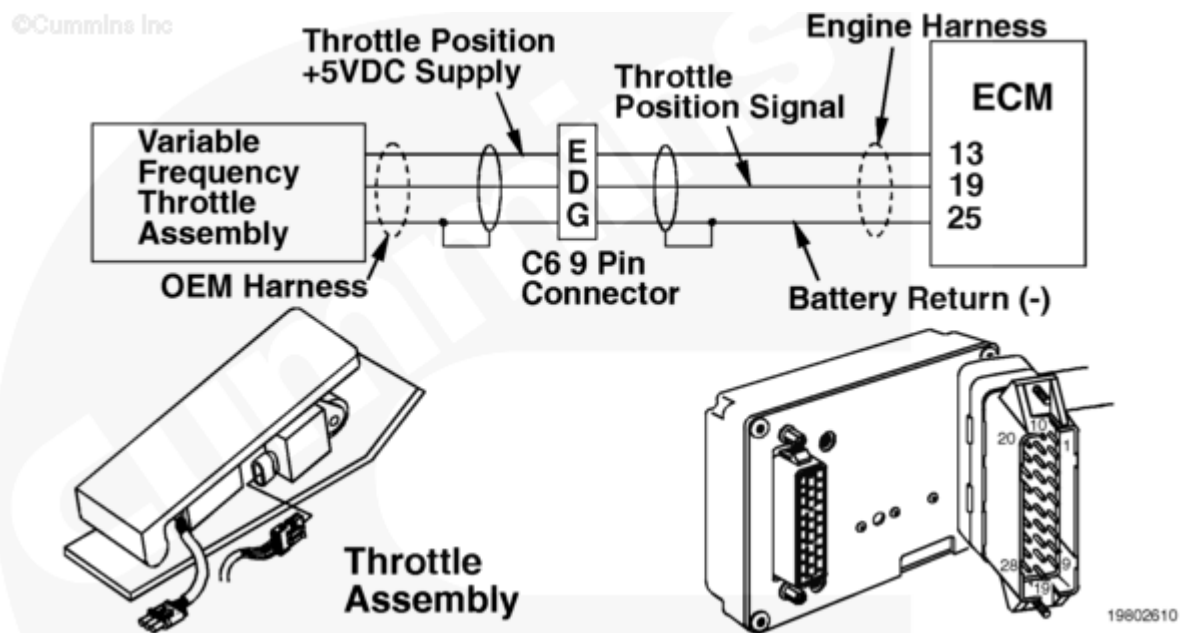
Fault Code: 443-fa

Accelerator Pedal or Lever Position Sensor Supply Voltage Circuit - Voltage Below Normal or Shorted to Low Source

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Overview

Codes	Reason	Effect
Fault Code: 443-fa PID(P): S254 SPN: FMI: 4 Lamp: Flashing SRT: 00-637	Low voltage detected at the sensor supply for the frequency throttle signal circuit.	Possible reduced engine performance.



Frequency Throttle Signal Circuit

Circuit Description

The throttle position sensor is a variable frequency sensor used by the ECM to monitor the throttle command. This sensor has three circuits: 5 volt supply, return and signal circuits. The signal circuit voltage indicates the position of the throttle position sensor.

Component Location

The variable frequency throttle assembly generator location varies with each OEM.

Conditions For Running The Diagnostics

This diagnostic runs continuously when the keyswitch is in the ON position.

Conditions For Setting The Fault Codes

The Engine Control Module (ECM) detected the sensor supply to the frequency throttle signal was out of range low.

Action Taken When The Fault Code Is Active

- The ECM illuminates the amber CHECK ENGINE light when the diagnostic runs and fails.

For Power Generation Applications:

- The generator set controller displays the fault immediately when the diagnostics runs and fails.

Conditions For Clearing The Fault Code

- To validate the repair, perform a key cycle, start the engine and let it idle for 1 minute.
- The fault code status displayed by the recommended Cummins® electronic service tool or equivalent will change to INACTIVE immediately after the diagnostic runs and passes.
- The ECM will turn off the amber CHECK ENGINE lamp immediately after the diagnostic runs and passes.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active and inactive faults.

For Power Generation Applications:

- To validate the repair, start the engine and let it run for 1 minute at no load.
- The generator set controller will turn off the indicator immediately after the user presses reset.
- The “Reset All Faults” command in the recommended Cummins® electronic service tool or equivalent can be used to clear active faults.

Shoptalk

Possible causes of this fault code include:

- Malfunctioning or damaged variable frequency throttle assembly.
- Malfunctioning or damaged engine wiring harness.

- Damaged or loose connectors.

Refer to Troubleshooting Fault Code t05-443 (</qs3/pubsys2/xml/en/procedures/98/98-t05-443fa.html>)

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